MASS SPECTROMETRY

1. **Sample introduction**
2. **Methods of sample ionization**
3. **Analysis and separation of sample ions**
4. **Detection and recording of sample ions.**

INLET SYSTEM	**ION SOURCE**	**SEPARATION OF IONS**	**DETECTION OF IONS**	**RECORDING OF ION**

1. Sample introduction

- The method of sample introduction to the ionisation source often depends on the ionisation method being used, as well as the type and complexity of the sample.
- The sample can be inserted directly into the ionisation source, or can undergo some type of chromatography *en route* to the ionisation source. This latter method of sample introduction usually involves the mass spectrometer being coupled directly to a high pressure liquid chromatography (HPLC), gas chromatography (GC) or capillary electrophoresis (CE) separation column, and hence the sample is separated into a series of components which then enter the mass spectrometer sequentially for individual analysis.

(EI Source)

- At pressure not more than 10^{-5} torr to 10^{-7} torr to avoid:
- 1) the collision of the ions with each other and with the molecules, as this would produce inter-ionic intermolecular or ion molecular reactions,
- ii) the erosion of the electron producing filament.
- *Methods:*
- i)-a) Direct insertion (Direct probe inlet)

  The sample is placed on a probe, which is then inserted into the ionization region of the mass spectrometer, typically through a vacuum interlock. The sample is either heated to facilitate thermal desorption or subjected to any number of high-energy desorption processes, such as laser desorption, to facilitate vaporization and ionization.
- i)-b) in-beam or direct ionization
- Chromatographic eluent insertionI
- Molecular Leak (cold inlet) for highly volatile solids and liquids and gaseous sample

2. Methods of sample ionization

The ionisation method to be used should depend on the type of sample under investigation and the mass spectrometer available.

Major Ionisation methods include the following:

Electron Impact (EI)
Chemical Ionisation (CI)
Electrospray Ionisation (ESI)
Fast Atom Bombardment (FAB)
Field Desorption / Field Ionisation (FD/FI)
Matrix Assisted Laser Desorption Ionisation (MALDI)
Thermospray Ionisation (TSP)

The ionisation methods used for the majority of biochemical analyses are **Electrospray Ionisation (ESI)** and **Matrix Assisted Laser Desorption Ionisation (MALDI)**

- *Electron Ionization.* Electron Ionization (EI) is the most common ionization technique used for mass spectrometry. EI works well for many gas phase molecules, but it does have some limitations. Although the mass spectra are very reproducible and are widely used for spectral libraries, EI causes extensive fragmentation so that the molecular ion is not observed for many compounds. Fragmentation is useful because it provides structural information for interpreting unknown spectra.

- The electrons used for ionization are produced by passing a current through a wire filament. (The amount of current controls the number of electrons emitted by the filament). An electric field accelerates these electrons across the source region to produce a beam of high energy electrons. When an analyte molecule passes through this electron beam, a valence shell electron can be removed from the molecule to produce an ion.

Electron Ionization Source

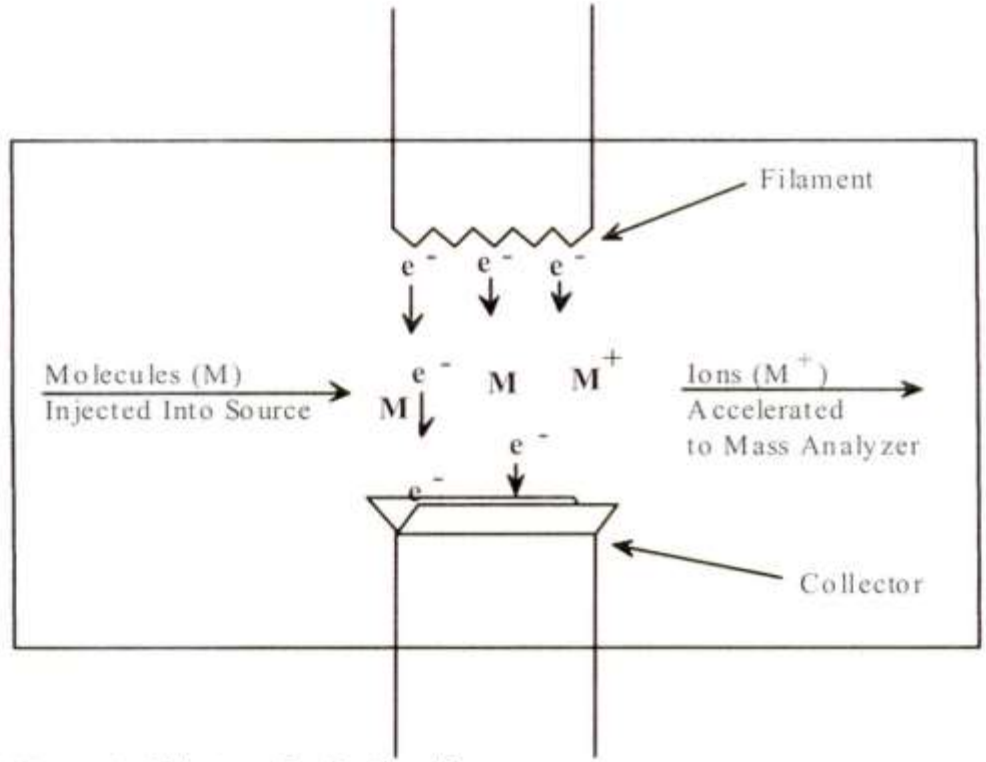


Figure 2. Electron Ionization Source.

- Ionization does not occur by electron capture, which is highly dependent upon molecular structure. Instead, EI produces positive ions by knocking a valence electron off the analyte molecule. As the electron passes close to the molecule the negative charge of the electron repels and distorts the electron cloud surrounding the molecule. This distortion transfers kinetic energy from the fast-moving electron to the electron cloud of the molecule. If enough energy is transferred by the process, the molecule will eject a valence electron and form a radical cation ($M^{+\cdot}$).

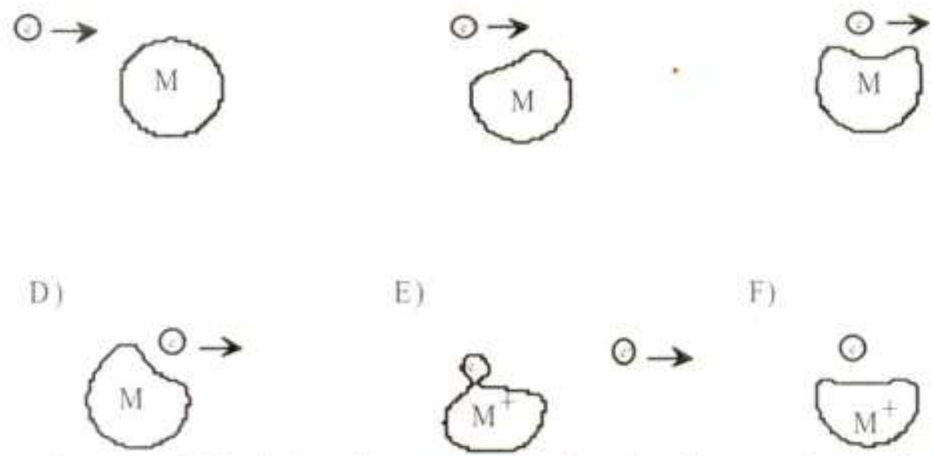


Figure 3 Electron Ionization Process. A) Ionizing electron approaches the electron cloud of a molecule; B) Electron cloud distorted by ionizing electron; C) Electron cloud further distorted by ionizing electron; D) Ionizing electron passes by the molecule; E) Electron cloud of molecule ejecting an electron; F) Molecular ion and ejected electron.

- Since the ionization is produced by a single electron that is accelerated to 70 V, this is commonly referred to as 70 eV EI. This is enough energy to cause extensive fragmentation. The amount of energy transferred during this process depends upon how fast the electron is traveling and how close it passes to the molecule. In most 70 eV EI experiments, approximately 15 eV of energy is transferred during the ionization process. There is, however, a distribution of energy and as much as 30 eV is transferred to some molecules. Since approximately 10 eV is required to ionize most organic compounds and a typical chemical bond energy is 3 eV, extensive fragmentation is often observed in 70 eV EI mass spectra. The distribution of energy transferred during ionization and the large number of fragmentation pathways results in a variety of products for a given molecule. Other electron voltages may be used to vary the amount of fragmentation produced during ionization. For most organic compounds the threshold energy for EI is about 10 eV. Because a mass spectrum is produced by ionizing many molecules, the spectrum is a distribution of the possible product ions. Intact molecular ions are observed from ions produced with little excess energy. Other molecular ions have more energy and undergo fragmentation in the source region. Changing the ionization energy changes the observed distribution of fragment ions. This distribution provides the structural information for interpreting mass spectra.

IONIZATION TECHNIQUES

***Hard ionization technique*:**

1. Electrons (EI)

(If E < 10eV ie < the IP of the molecule no apparent change takes place).

$$M + e \xrightarrow{E\ 70 \cong eV} M^{+} + 2e$$

(The ions spend 10^{-6} s after formation in the ion chamber)

SOFT IONIZATION TECHNIQUES

- ***Fast Atom Bombardment and Secondary Ion Mass Spectrometry***

Fast Atom Bombardment (FAB) and Secondary Ion Mass Spectrometry (SIMS) both use high energy atoms/ions to sputter and ionize the sample in a single step. In these techniques, a beam of rare gas neutrals (FAB) or ions (SIMS) is focused on the liquid or solid sample. The impact of this high energy beam causes the analyte molecules to sputter into the gas phase and ionize in a single step (Figure 6). The exact mechanism of this process is not well understood, but these techniques work well for compounds with molecular weights up to a few thousand dalton. Since no heating is required, sputtering techniques (especially FAB) are useful for studying thermally labile compounds that decompose in conventional inlets.

- The most significant difference between FAB and SIMS is the sample preparation. In FAB the analyte is dissolved in a liquid matrix. A drop of the sample/matrix mixture is placed at the end of an insertion probe and introduced to the source region. The fast atom beam is focused on this droplet to produce analyte ions. Glycerol or similar low vapor pressure liquids are typically used for the matrix. Ideally, the analyte is soluble in the liquid matrix and a monolayer of analyte forms on the surface of the droplet. According to one theory, this monolayer concentrates the analyte while the dissolved sample provides a reservoir to replenish the monolayer as the analyte is depleted. Without this constant replenishment from the bulk solution, the ionizing beam will rapidly deplete the analyte and the signal is difficult to observe.

- SIMS experiments are used to study surface species and solid samples. No matrix is used and the ionizing beam is focused directly on the sample. Although this makes sampling more difficult, it is useful for studying surface chemistry. High resolution chemical maps are produced by scanning a tightly focused ionizing beam across the surface and depth profiles are produced by probing a single location(*9*, *10*). Although SIMS is a very sensitive and powerful technique for surface chemistry and materials analysis, the results are often difficult to quantitate.

Fast atom Bombardment (FAB) Mass Spectrometry

High energy particles
To desorb charged molecular species from the surface of a liquid in which it is dissolved.
The ions in the gaseous phase are analysed in the usual manner.

Eg:
Xenon atoms ⟶ ionized $Xe^{+\cdot}$ **Ions** ⟶
Accelerated Fast Ions

$Xe^{+\cdot}$ + Xe (thermal) ⟶ Xe $Xe^{+\cdot}$
(Fast) (gas chamber) (Fast) (Thermal)
access is
deflected.

Beam of fast atom bombards a metal plate coated with the sample.

The large amount of K.E. in the atoms volatilizes and ionizes the sample.

Either-ve or +ve ions may be produced and analyzed.

Other gases: He, Ar.

In-volatile polar liquids e.g. glycerol, triethyl amine, 1,1,3,3-tetramethyl urea, thioglycerol, crown ethers.

Desorption takes place by a combination of:

i) **Thermal Vaporization:**
due to the localized heating caused by the impact of a single H.E. particle.

ii) **Sputtering:**
Molecules interact with the impacting particles as hard spheres and desorption occurs during a very short time interval following impact and prior to equilibration of energy.

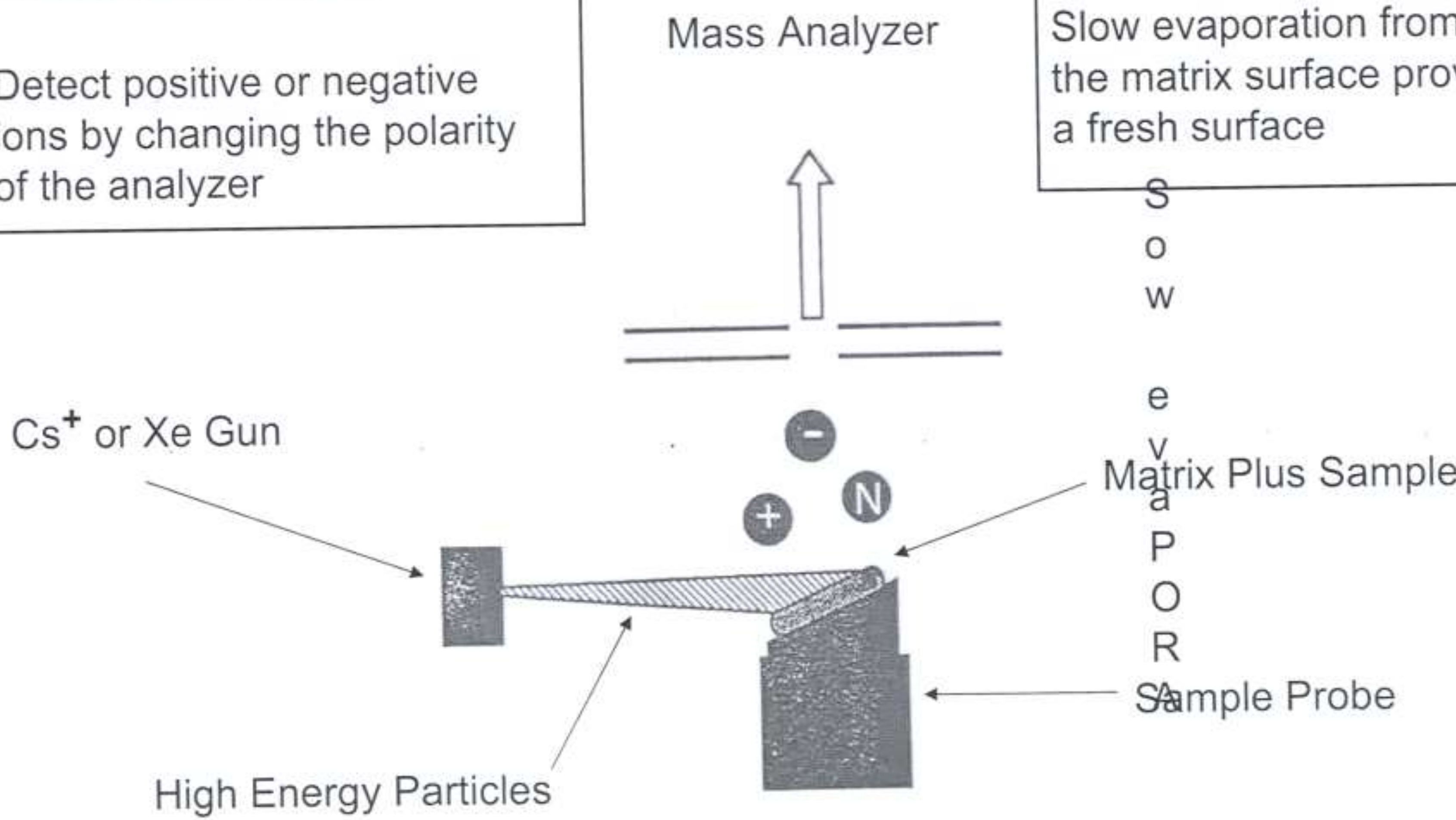
FAB/LSIMS Ion Source

High energy atoms or ions desorb positive ions, negative ions and neutrals from the surface of the matrix.

Detect positive or negative ions by changing the polarity of the analyzer

Use scanning analyzers such as magnetic sector or quadrupole

Slow evaporation from the matrix surface provides a fresh surface



Common Matrices

Glycerol MW 92

1-S Glycerol MW 108

m-Nitrobenzyl Alcohol MW 153

Sample must be soluble
In the matrix

Matrix must have low volatility

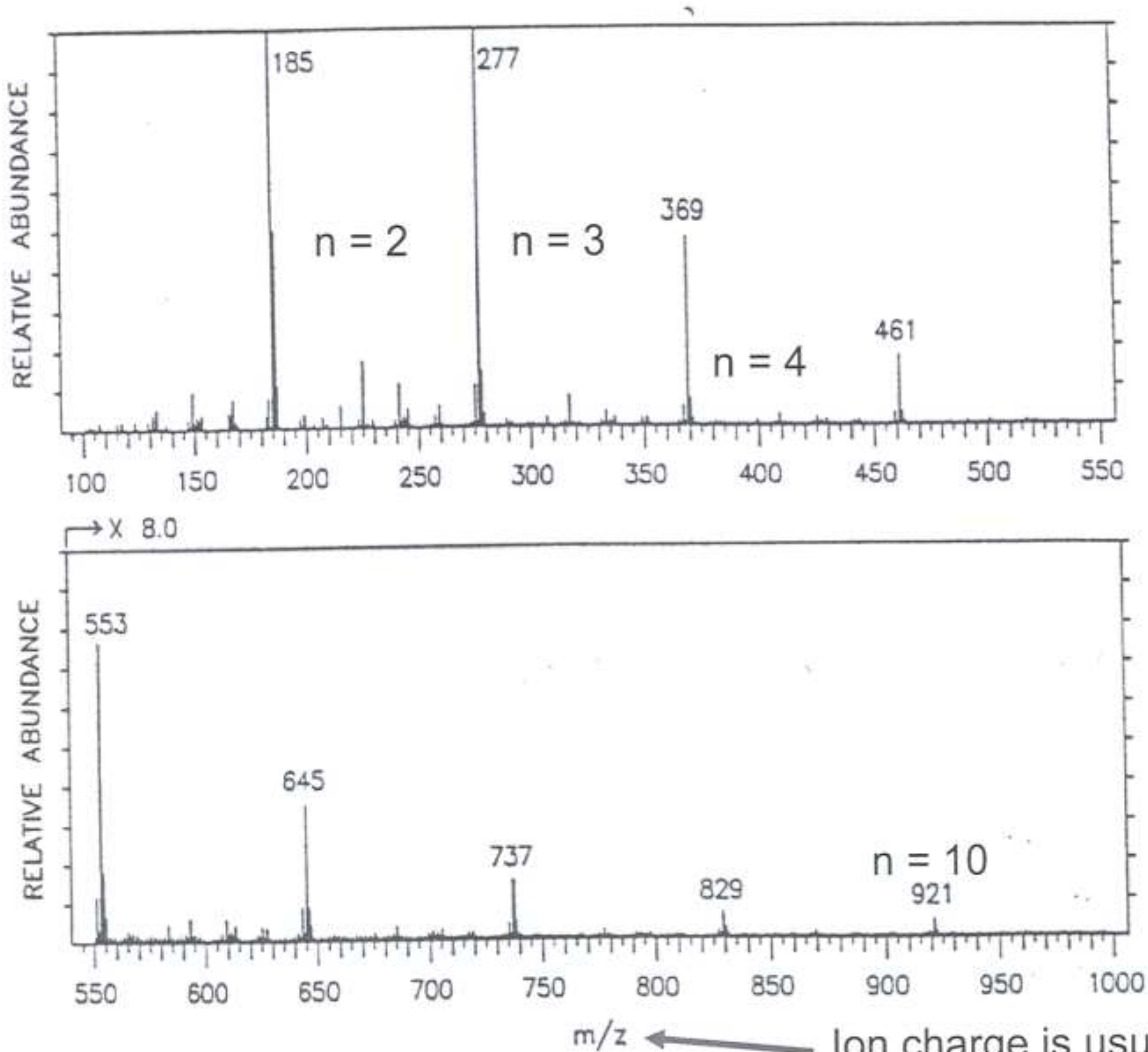
Despite much effort in several laboratories, substantially better matrices have not been reported

Additives can be used to enhance spectra

Most effective for non-ionic polar compound 0.1-1% by weight usually works well

Additive	**Pos Ion**	**Neg Ion**
NaCl	MNa^+	MCl^-
HCl	MH+	MCl^-
NH_4OH	MNH_4^+	$M\text{-}H^-$

Positive Ion FAB Spectrum of Glycerol MW 92



Ion charge is usually 1

Major Peaks G_nH^+

n = 1, 2, 3, 4 …
H = Proton

High levels of salt give extra ions

$G_nNa^+ \Rightarrow +23$
Na At Wt = 23
$G_nNa^+ - G_nH^+ = 22$

$G_nK^+ \Rightarrow +39$
K At Wt = 39
$G_nK^+ - G_nH^+ = 38$

Main Features of FAB:

High molecular wt., low volatility subs. e.g. oligonucleotides containing upto 10 nucleotides.

$(M^{+}+H)$ as well as adduct and frag. Ions

(Like in CI)

$(M^{+}-H)$ ions

Peptides upto 3,000 daltons

Complete amino acid sequence from the key fragment ions

Phospholipids

Carbohydrates

Glycosides

Chemical Ionization

For organic chemists, Chemical Ionization (CI) is especially useful technique when no molecular ion is observed in EI mass spectrum, and also in the case of confirming the mass to charge ratio of the molecular ion. Chemical ionization technique uses virtually the same ion source device as in electron impact, except, CI uses tight ion source, and reagent gas. Reagent gas (e.g. ammonia) is first subjected to electron impact. Sample ions are formed by the interaction of reagent gas ions and sample molecules. This phenomenon is called ion-molecule reactions. Reagent gas molecules are present in the ratio of about 100:1 with respect to sample molecules. Positive ions and negative ions are formed in the CI process. Depending on the setup of the instrument (source voltages, detector, etc...) only positive ions or only negative ions are recorded.

Unlike molecular ions obtained in EI method, MH^+ and $[M-H]^-$ detection occurs in high yield and less fragment ions are observed.

Chemical Ionization (CI) is a soft ionization technique that produces ions with little excess energy. As a result, less fragmentation is observed in the mass spectrum. Since this increases the abundance of the molecular ion, the technique is complimentary to 70 eV EI. CI is often used to verify the molecular mass of an unknown. Only slight modifications of an EI source region are required for CI experiments. In Chemical Ionization the source is enclosed in a small cell with openings for the electron beam, the reagent gas and the sample. The reagent gas is added to this cell at approximately 10 Pa (0.1 torr) pressure. This is higher than the 10^{-3} Pa (10^{-5} torr) pressure typical for a mass spectrometer source. At 10^{-3} Pa the mean free path between collisions is approximately 2 meters and ion-molecule reactions are unlikely. In the CI source, however, the mean free path between collisions is only 10^{-4} meters and analyte molecules undergo many collisions with the reagent gas. The reagent gas in the CI source is ionized with an electron beam to produce a cloud of ions. The reagent gas ions in this cloud react and produce adduct ions like CH_5^+, which are excellent proton donors.

When analyte molecules (M) are introduced to a source region with this cloud of ions, the reagent gas ions donate a proton to the analyte molecule and produce MH^+ ions. The most common reagent gases are methane, isobutane and ammonia. Methane is the strongest proton donor commonly used with a proton affinity (PA) of 5.7 eV. For softer ionization, isobutane (PA 8.5 eV) and ammonia (PA 9.0 eV) are frequently used. Acid base chemistry is frequently used to describe the chemical ionization reactions. The reagent gas must be a strong enough Brønsted acid to transfer a proton to the analyte. Fragmentation is minimized in CI by reducing the amount of excess energy produced by the reaction. Because the adduct ions have little excess energy and are relatively stable, CI is very useful for molecular mass determination.

Chemical Ionization (CI)

- Chemical ionization is a lower energy process than electron ionization (EI). The lower energy yields less or sometimes no fragmentation, and usually a simpler spectrum.
- Softer ionization technique →Less fragmentation → Easier to find molecular ions.
- Two different modes: **Negative chemical ionization (NCI)** and **Positive Chemical Ionization (PCI)**.

Positive chemical ionization:

Methane:

CH_4 + e -----> $CH_4^{+\cdot}$ + 2e

$CH4^{+\cdot}$ + CH_4 -----> CH_5^{+} + $CH_3^{\cdot}$

$CH_4^{+\cdot}$ + CH_4 -----> $C_2H_5^{+}$ + H_2 + $H^{\cdot}$

Isobutane:

i-C_4H_{10} + e -----> i-$C_4H_{10}^{+\cdot}$ + 2e

i-$C_4H_{10}^{+\cdot}$ + i-C_4H_{10} ------> i-$C_4H_9^{+}$ + C_4H_9 +H_2

Ammonia:

NH_3 + e -----> $NH_3^{+\cdot}$ + 2e

$NH_3^{+\cdot}$ + NH_3 ------> NH_4^{+} + $NH_2^{\cdot}$

NH_4^{+} + NH_3 ---------> $N_2H_7^{+}$

In methane positive ion mode CI the relevant peaks observed are MH^{+}, $[M+CH_5]^{+}$, and $[M+C_2H_5]^{+}$; but mainly MH^{+}

In isobutane positive ion mode CI the main peak observed is $MH^{+\cdot}$

In ammonia positive ion mode CI the main peaks observed are MH^{+} and $[M+NH_4]^{+\cdot}$

Choice of reagent gas:

Two factors determine the choice of the gas to be used:

- **Proton affinity (PA): $H^+ + M$ -------MH^+**
- **(Energy transfer)**

•NH_3 (ammonia) is the most used reagent gas in CI because of the low energy transfer of NH_4^+ compare to CH_5^+ for example. With NH_3 as reagent gas, usually MH^+ and MNH_4^+ (17 mass units difference) are observed.

Positive ion mode:

- A "tight" ion source (pressure=0.1-2 torr) is used to maximize collisions which results in increasing sensitivity.

- Proton transfer is one of the simple processes observed in positive CI:

$RH^+ + M \text{ -----> } MH^+ + R$

- **One of the decisive parameter in this reaction is the proton affinity. For the reaction to occur, the proton affinity of the molecule M must be higher than that of the gas molecule.**
- Choice of reagent gas affect the extent of fragmentation of the quasi-molecular ion.

Excitation energy of MH^+ = PA (M) –PA (R)

•**Excitation energy in MH^+ facilitates fragmentation**

Table 1. Reagent species and their properties for the common PCI gases.

Reagent Gas	Major Species	CI Products	Proton Affinity (kcal/mole)
Methane	CH_5^+	M+1	131.6
	$C_2H_5^+$	M+29	162.6
	$C_3H_5^+$	M+41	–
Isobutane	$C_4H_9^+$	M+1	195.9
Ammonia	NH_4^+	M+1	204.0
		M+18	–

CHEMICAL IONIZATION:

$CH_4 + e \longrightarrow CH_4^{+\cdot} + 2e$ [~ 10^{-6} tor; 70 eV] (in EI)

$CH_4 + CH_4^{+\cdot} \longrightarrow CH_5^{+} + CH_3^{\cdot}$

Reagent gas → ion chamber ← Sample gas

1Torr; eV 100-500

ion chamber

Sample gas

10^{-3} Torr

R + e → $[R^{+\cdot}]^{*}$ excited primary reactant gas ions → collisional H-transfer → (R+H) equilibrated sec reactant gas ion

.

$(R+H)^+$ + M $\longrightarrow$ $(M+H)^+ + R$

reactant gas ion

molecule under investigation

$(M-H)^+$ *quasi*-mol (tertiary) ions

$+RH_2$ neutral species

CH_4	affords	principally	CH_5^+
Isobutane	=	=	$C_4H_9^+$
NH_3	=	=	NH_4^+
H_2	=	=	H_3^+

R:M should be atleast 10^3 at a 1 tor

Basics of (Positive-Ion) Chemical Ionization

In positive-ion chemical ionization (PICI) new ionized species are formed when gaseous molecules interact with ions. CI may involve the transfer of an electron, proton or other charged species between the reactants.
The reactants are:
i) the neutral analyte M
ii) ions from a reagent gas (methane, isobutane, ammonia, etc.)
Bimolecular processes are used to generate analyte ions. The occurrence of bimolecular reactions requires a sufficiently large number of ionmolecule collisions in the ion source, i.e. a non negligible partial pressure of the reagent gas.
Assuming reasonable collision cross sections and an ion source residence time of 1 μs a molecule will undergo 30–70 collisions at an ion source pressure of about 2.5 × 102 Pa = 0.002 atm = 1.9 torr.
A 103–104-fold excess of reagent gas is then required; such an excess is also useful to shield the analyte molecules from ionizing primary electrons i.e. to suppress competing direct EI of the analyte.

Four general pathways are hypothesized to form ions from a neutral analyte M in (positive) CI:

$M + [BH]^+$	$\longrightarrow [M+H]^+ + B$	proton transfer
$M + X^+$	$\longrightarrow [M+X]^+$	electrophilic addition
$M + X^+$	$\longrightarrow [M-A]^+ + AX$	anion abstraction
$M + X^{+\cdot}$	$\longrightarrow M^{+\cdot} + X$	charge exchange

1. A competitive generation of [M-H]- ions, due to proton abstraction, has to be considered in the case of acidic analytes

2. Electrophilic addition is typical of NH4 + ions when ammonia is used as reagent gas

3. Hydride abstractions are common examples of anion abstraction, typical for aliphatic alcohols

4. Charge exchange yields radical ions with low internal energy, which behave similarly to molecular ions in low-energy EI.

Negative Ion Chemical Ionization:

In order to see a response by negative chemical ionization (NCI), the analyte must be capable of producing a negative ion (stabilize a negative charge).

NCI can be used for the analysis of compounds containing acidic groups or electronegative elements (especially halogens)

Three mechanisms can be underlined:

1- Electron capture reaction due to attainment of slow moving, low energy "thermalized" electrons which may be transfered more efficiently to sample molecules.

$$A + e^{-} \qquad\qquad A^{-}$$

- **2. Electron transfer from ionized reagent gas (e.g. NH_2^- may transfer an electron to a molecule having a greater electron affinity than NH_2).**

- **3. Reagent gas ions participation in true CI reactions (e.g. proton abstraction, according to relative acidities).**

Molecular ions observed in negative ion chemical ionization mass spectra are usually M^- or $[M\text{-}H]^-$

- Summary of common chemical ionization methods
  - Chemical ionization (CI)

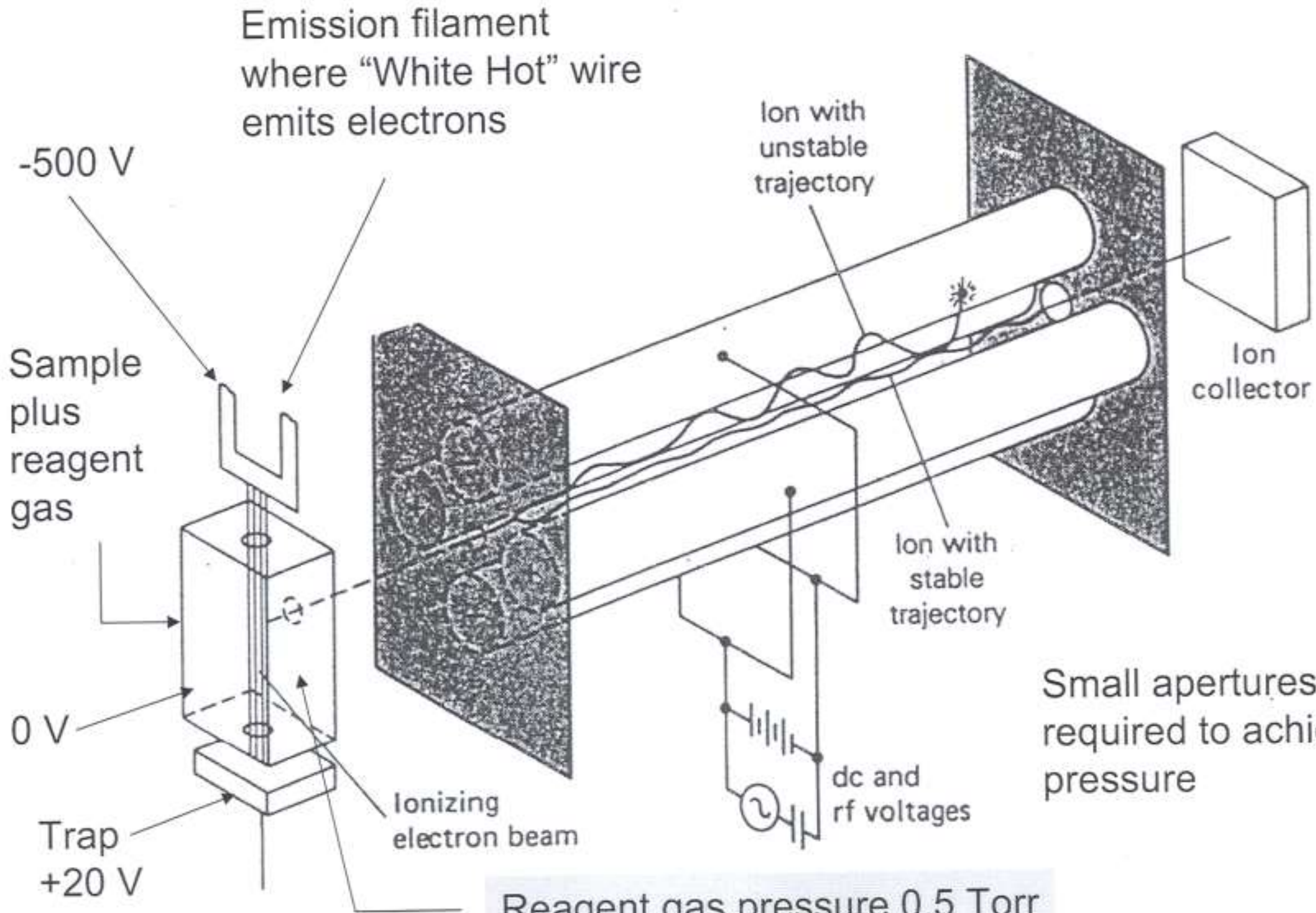
- Positive-Ion CI (PICI):
- Proton transfer
- Charge exchange
- Electrophilic addition
- Anion abstraction

Negative-Ion CI (NICI):

Electron capture (EC)

Ion-molecule reactions

Chemical Ionization (CI)



Small apertures are required to achieve high pressure

Reagent gas pressure 0.5 Torr

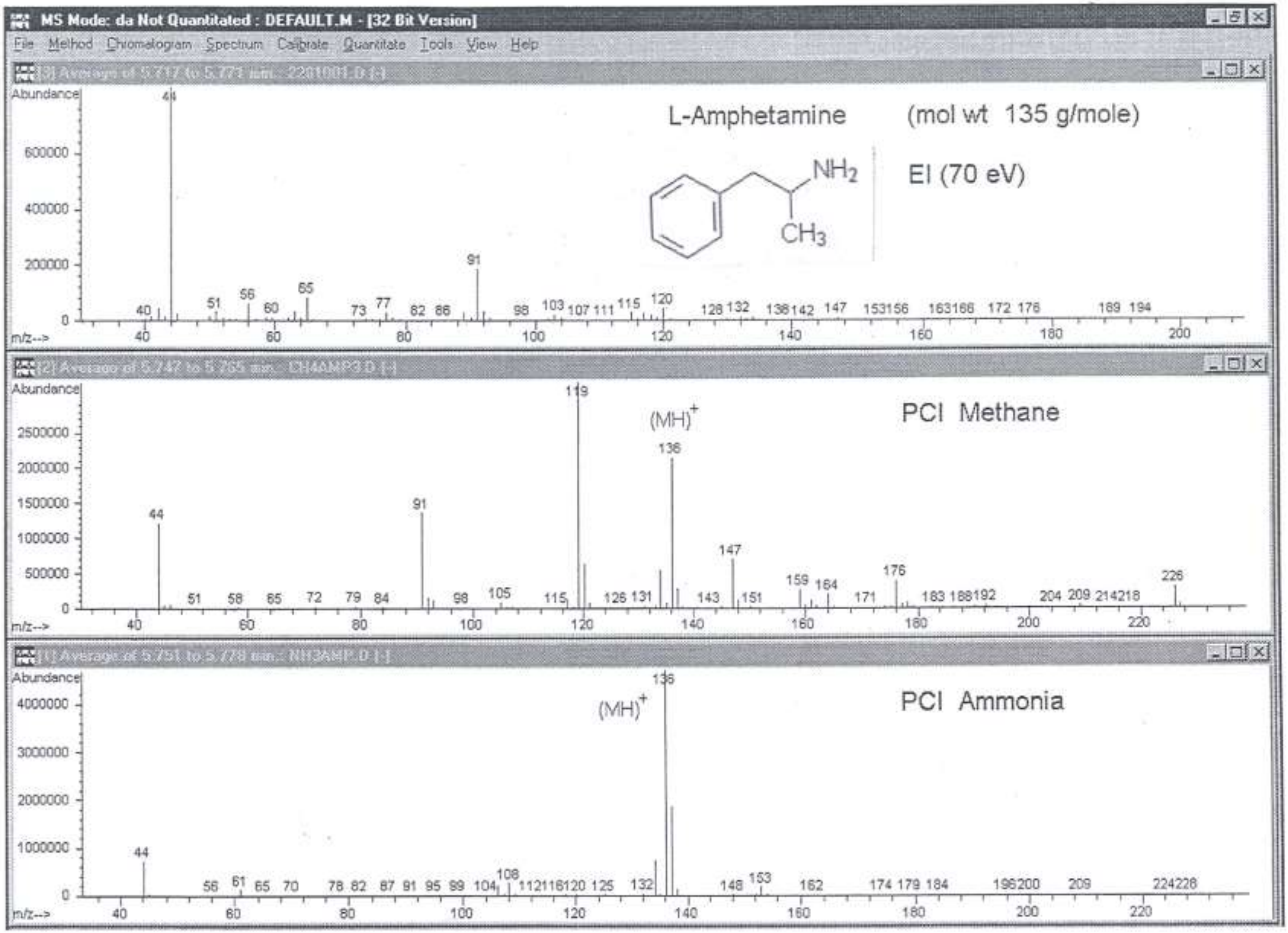


Figure 1. Mass spectrum of amphetamine (molecular weight 135 g/mole) as obtained by EI at 70-eV (upper panel), and PCI with methane (middle panel) and with ammonia (lower panel) reagent gas. Notice the decreasing degree of fragmentation most clearly indicated by the decreasing abundance of the 44 *m/z* fragment.

SUMMARY (SECOND LECTURE)

- Sample introduction
- Methods of sample ionization
- Comparison b/w CI and EI; source modifications required to increase pressure
- Why is high reagent gas pressure required?
- Reagent gas; common gases; reagent ions; common reactions
- Proton transfer; hydride abstraction; adduct formation
- Proton Affinity; how does it relate to ion excitation energy and fragmentation
- Use proton affinity to predict PT or A in ammonia CI
- Explain apparent M+ in ammonia CI spectra
- Fast Atom Bombardment Ionization (FAB) and Liquid Secondary Ion Mass Spectrometry (LSIMS)